EE 474 Note on Linearization

Contents

1 Plant modeling and linearization **1**
1.1 A note on capturing the plant model 1
1.2 Linearization of a nonlinear plant 1

1 Plant modeling and linearization

Modeling is an important part of control system analysis and realization. Obtaining accurate model parameters is an important part of the cost of practical control systems. In this section, some broad modeling concepts are introduced that will be important for working with the SRV02 motor servo throughout the semester.

1.1 A note on capturing the plant model

It may seem as if the dynamic model of a system, such as $G_{uv}(s)$ for the SRV02, should be fixed, capturing all available knowledge of the dynamics of the system. But in practical cases the model complexity varies, depending on the extent of available knowledge and budget for determining accurate parameters. In an oil refinery with 10,000 control points - valves, pumps, heating units - the budget may restrict modeling for each control point to simple models, perhaps as simple as Eqn (1), below. In contrast, in large aerospace programs, such as the Boeing 787 or F-35, where the system is complex and many identical units will be built, and where a small difference in performance can bring significant value, tens of millions of dollars and hundreds of person-years of effort may be spent developing a detailed dynamic model.

During the EE 474 lab sequence, you will be working extensively with the model of the SRV02, starting with a very simple model and progressively developing more detail. In this lab, you will work with the simplest model,

$$G_{uv}(s) = K_{uv} \tag{1}$$

where $G_{uv}(s)$ is the transfer function of the motor servo, and K_{uv} [radians/sec/volt] is a constant that relates the applied voltage to the *steady-state* rotation rate.

1.2 Linearization of a nonlinear plant

Plant is a name often given to the process or system being controlled. A single-input, single-output plant is illustrated in figure 1.

The plant of figure 1 is shown with governing equation

$$y(t) = f(u(t)) \,. \tag{2}$$

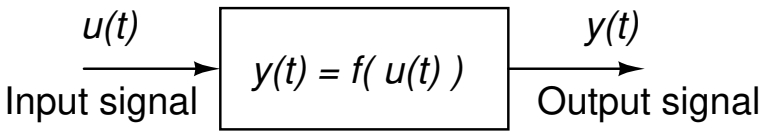


Figure 1: Plant with input and output signals.

For an ideal ***linear*** system, the governing equation can be simplified to a linear equation, such as

$$y(t) = K_{uv}\, u(t) \tag{3}$$

where K_{uv} is a constant gain relating the input to the output.

However, many systems, including the SRV02 motor servo we use in lab, are governed at least partially by nonlinear effects. In the SRV02, the main nonlinearity is friction. A possible non-linear input-output relationship is seen in figure 2. To work with nonlinear system elements and apply linear system theory, we ***linearize*** the response curve of the system element about an ***operating point***.

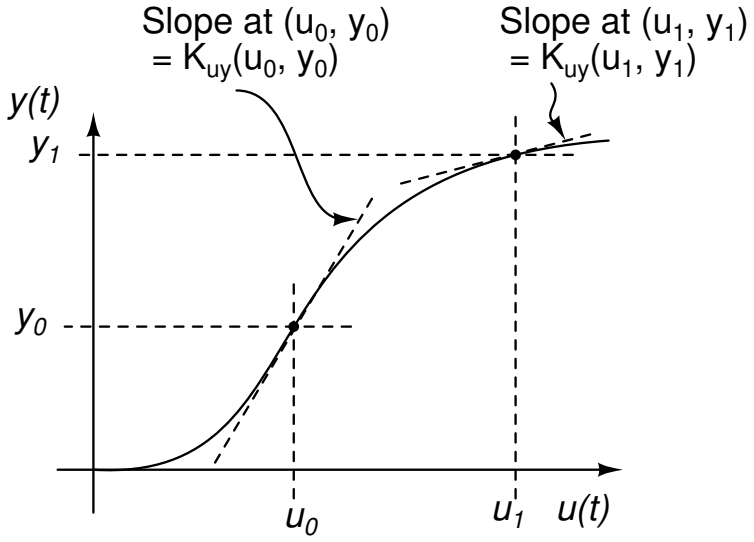


Figure 2: Possible non-linear input-output relationship of a system element (this curve might be typical of a valve, for example an engine throttle). Two operating points are shown.

Linearization about an operating point involves two parts:

1. The operating point, which is the level on the input-output curve about which the system will be linearized. The operating point can be expressed as an input-output pair (u_0, y_0).

2. The slope of the input-output curve at the operating point, which is the gain of the system element at that operating point. The gain at the operating point can be expressed as the gain $K_{uv}(u_0, y_0)$.

Starting with the nonlinear input-output relationship

$$y(t) = f(u(t)) \tag{4}$$

and an operating point (u_0, y_0) and expressing the input and output signals as deviations about the operating point,

$$u(t) = u_0 + \delta u(t) , \qquad y(t) = y_0 + \delta y(t) \tag{5}$$

and with $K_{uv}(u_0, y_0)$ as the slope of the input-output curve at (u_0, y_0), then the deviation in the output is given from the deviation in the input by:

$$\delta y(t) \simeq K_{uv}(u_0, y_0)\, \delta u(t) \tag{6}$$

Equation (6) is a *linear* relationship between the input and output *deviations*. Putting Equations (5) and (6) together, we may write:

$$y(t) \simeq y_0 + K_{uv}(u_0, y_0)\, \delta u(t) \tag{7}$$

Equations (6) and (7) show three important details:

1. The linearization applies to deviations about the operating point,
2. The gain depends on the operating point, and
3. The linearization is approximate.

However, linearization is so widely used in control system analysis and design that a shorthand is often used. Starting with the equation in small deviations,

$$\delta y(t) \simeq K_{uv}(u_0, y_0)\, \delta u(t) \qquad \text{(Eqn (6), repeated)}$$

the linearized input-output relationship is written as:

$$y(t) = K_{uv}\, u(t) \tag{8}$$

where the shorthand is to omit the details of the operating point, and using the $y(t)$ and $u(t)$ symbols to represent the deviations from the operating point, rather than the absolute magnitude of the signals.

Equation (8) is quicker to write than equations (5), (6) or (7), and so in control system analysis and design, the input-output relationship of a linearized system element is very often written as equation (8); but we must remember that when there has been a linearization, the three important details apply and equations of the form of equations (5), (6) and (7) are the true description of the system.

Example :

For an aircraft autopilot set to cruise at an altitude of 10,000 meters, the altitude signal, $h(t)$, will be the deviation from 10,000 meters, actually $\delta h(t)$ in the notation of Eqn (5). Using common shorthand however, symbol $h(t)$ may be spoken of as the altitude, but actually refers to the *deviation* from the nominal altitude, h_O .